

Encrypt / Decrypt using Pen & Paper Method:

① Encrypt "CHASE YOUR DREAMS"

a) Caesar Cipher ($n=3$)

$$C = (P+3) \bmod 26$$

Plaintext	$P+3$	$(P+3) \bmod 26$	Ciphertext
C	5	5	F
H	10	10	K
A	3	3	D
S	21	21	V
E	7	7	H
Y	27	1	B
O	17	17	R
U	23	23	X
R	20	20	U
D	6	6	G
R	20	20	U
E	7	7	H
A	3	3	D
M	15	15	P
S	21	21	V

Ciphertext: FKDVH BRXU GUHDPV

b) Shift Cipher with key 7

$$C = (P + 7) \bmod 26$$

Plaintext	$P+7$	$(P+7) \bmod 26$	Ciphertext
C	9	9	J
H	14	14	O
A	7	7	H
S	25	25	Z
E	11	11	L
Y	31	5	F
O	21	21	V
U	27	1	B
R	24	24	Y
D	10	10	K
R	24	24	Y
E	11	11	L
A	7	7	H
M	19	19	T
S	25	25	Z

Ciphertext: JOH2L FVBY KYLHTZ

c) ROT 13

Plaintext	$P+13$	$(P+13) \bmod 26$	Ciphertext
C	15	15	P
H	20	20	U
A	13	13	N
S	31	5	F
E	17	17	R
Y	37	11	L
O	27	1	B
U	33	7	H
R	30	4	E
D	16	16	Q
R	30	4	E
E	17	17	R
A	13	13	N
M	25	25	Z
S	31	5	F

Ciphertext = PUNER LBHE QERNZF

2) Decrypt

Q) WRWDOOR VHFUHW PHVVDJH

Cipher Text	$P(C-3)$	$(C-3) \bmod 26$	Plaintext
W = 22	19	19	T
R = 17	14	14	O
W = 22	19	19	T
D = 3	0	0	A
O = 14	11	11	L
O = 14	11	11	L
B = 1	-2	24	Y
V = 21	18	18	S
H = 7	4	4	E
F = 5	2	2	C
U = 20	17	17	R
H = 7	4	4	E
W = 22	19	19	T
P = 15	12	12	M
H = 7	4	4	E
V = 21	18	18	S
V = 21	18	18	S
D = 3	0	0	A
J = 9	6	6	G
H = 7	4	4	E

Plaintext = TOTALLY SECRET MESSAGE

b) M J Ø Ø T B T W Ø I Shift Cipher 5

Cipher Text	$C-5$	$(C-5) \bmod 26$	Plaintext
M = 12	7	7	H
J = 9	4	4	E
Ø = 16	11	11	L
Ø = 16	11	11	L
T = 19	14	14	O
B = 1	-4	22	W
T = 19	14	14	O
W = 22	17	17	R
Ø = 16	11	11	L
I = 8	3	3	D

Plaintext = HELLO WORLD



§ STB DTWP YNRT Shift Cipher 5

Ciphertext	C-5	(C-5) mod 26	Plaintext
S = 18	13	13	N
T = 9	4	4	E
B = 1	1 4	22	W
D = 3	2 2	24	Y
T = 19	14	14	O
W = 22	17	17	R
P = 15	10	10	K
Y = 24	19	19	T
N = 13	8	8	I
R = 17	12	12	M
T = 9	4	4	E

Plaintext = NEW YORK TIME

Q. "FRAQ ZBER PBSSRR" ROT13

Cipher Text	C-13	(C-13) mod 26	Plaintext
F = 5	-8	18	S
R = 17	4	4	E
A = 0	-13	13	N
Q = 16	3	3	D
Z = 25	12	12	M
B = 1	-12	14	O
E = 4	-9	17	R
R = 17	4	4	E
P = 15	2	2	C
B = 1	-12	14	O
S = 18	5	5	F
S = 18	5	5	F
R = 17	4	4	E
R = 17	4	4	E

Plaintext = SEND MORE COFFEE